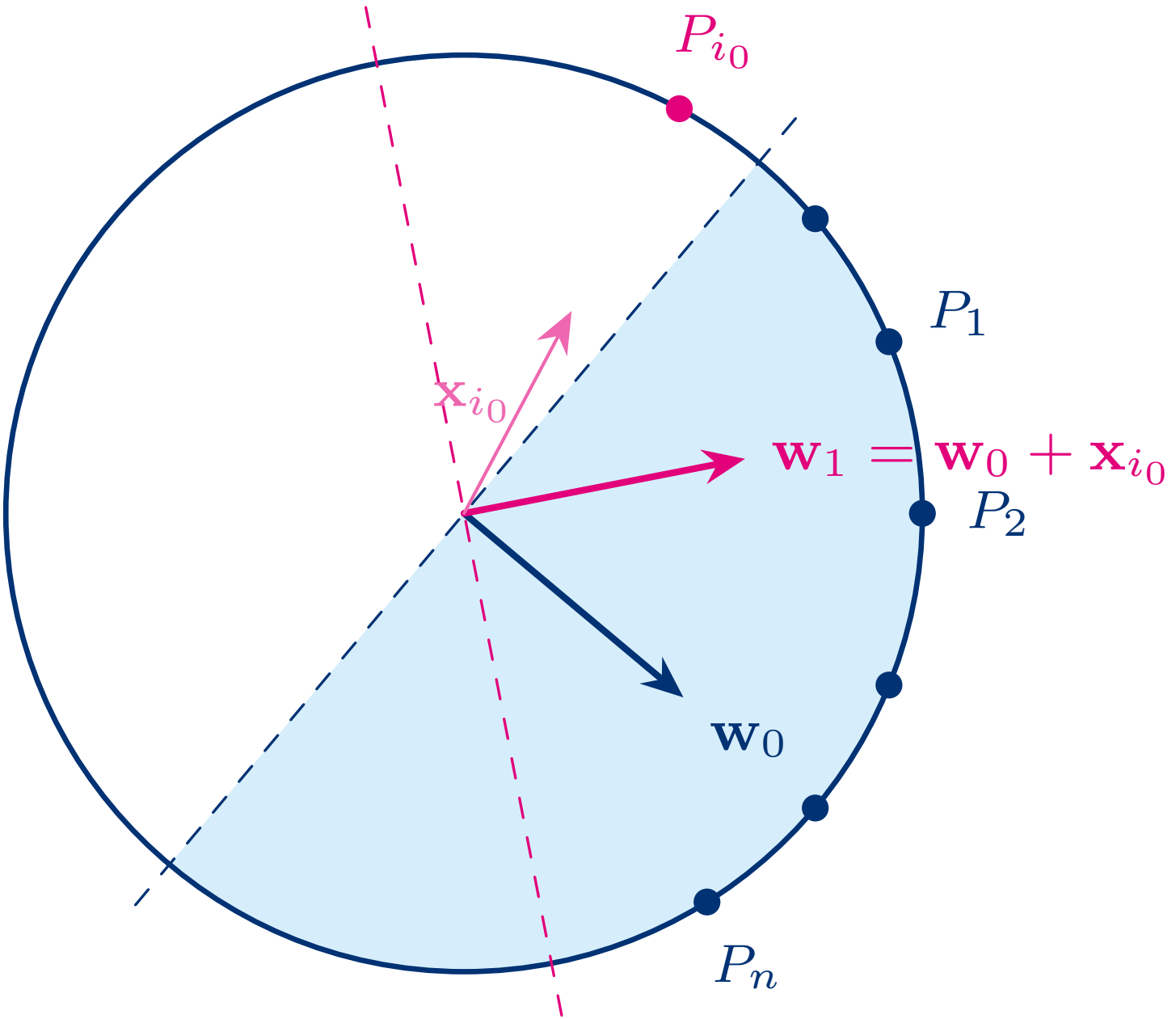

P_{i_0}
P_1
$\mathbf{x}_{i_0}$
$\mathbf{w}_1 = \mathbf{w}_0 + \mathbf{x}_{i_0}$
P_2
$\mathbf{w}_0$
P_n